Chapter 7 Practice: Exploring Personal Music Preferences

A Spotify API Analysis Using R

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1 Introduction

This project explores my personal music preferences using data retrieved from the Spotify Web API.

The objective is to investigate how my musical preferences differ across three listening periods and identify which artists remain consistently represented in my listening habits.

The analysis addresses three questions:

1. Who are my most preferred artists in the short term?
2. How do my artist rankings differ across listening periods?
3. Which artists remain consistent across all three periods?

2 Data Collection

2.1 Accessing the Spotify API

Data were collected from the Spotify Web API using the `httr` package in R.

After creating a Spotify Developer application, I configured authentication credentials and authorized access to my personal listening data.

I used the API's `/me/top/artists` endpoint to retrieve my top ten artists across three listening periods:

- **Short-term:** Approximately four weeks
- **Medium-term:** Approximately six months
- **Long-term:** Approximately one year

These rankings represent Spotify's calculated listening affinity rather than exact play counts.

The collected data were saved locally to make the analysis reproducible without requiring a new API request during every render.

2.2 Exploring the Dataset

```
head(my_spotify_data)
```

```
# A tibble: 6 x 3
   rank artist            period
  <int> <chr>             <fct>
1     1 Neil Young        Short-term
2     2 21 Savage         Short-term
3     3 Lil Baby          Short-term
4     4 Kodak Black       Short-term
5     5 Drake             Short-term
6     6 Alanis Morissette Short-term
```

The dataset contains three variables:

- `rank`: The artist's ranking within a listening period.
- `artist`: The name of the artist.
- `period`: The listening period associated with the ranking.

3 Data Preparation

3.1 Organizing Listening Periods

The data were separated into three datasets corresponding to Spotify's listening periods.

```
short_term <- my_spotify_data %>%
  filter(period == "Short-term")

medium_term <- my_spotify_data %>%
  filter(period == "Medium-term")

long_term <- my_spotify_data %>%
  filter(period == "Long-term")
```

The three periods are then summarized to examine the structure of the dataset.

```
my_spotify_data %>%
  count(period, name = "Number of Artists") %>%
  knitr::kable(
    caption = "Artist Rankings by Listening Period"
  )
```

Table 1: Artist Rankings by Listening Period

period	Number of Artists
Long-term	10
Medium-term	10
Short-term	10

4 Analysis and Visualization

4.1 My Top 10 Artists

The first visualization presents my top ten artists from the short-term listening period.

```
ggplot(
  short_term,
  aes(
    x = rank,
    y = forcats::fct_reorder(artist, -rank)
  )
) +
  geom_point(
```

```
    color = "#1DB954",
    size = 4
  ) +
  scale_x_reverse(
    breaks = 1:10
  ) +
  labs(
    title = "My Spotify Top 10 Artists",
    subtitle = "Short-term listening preferences",
    x = "Spotify Ranking (1 = Highest)",
    y = "Artist"
  ) +
  theme_minimal()
```

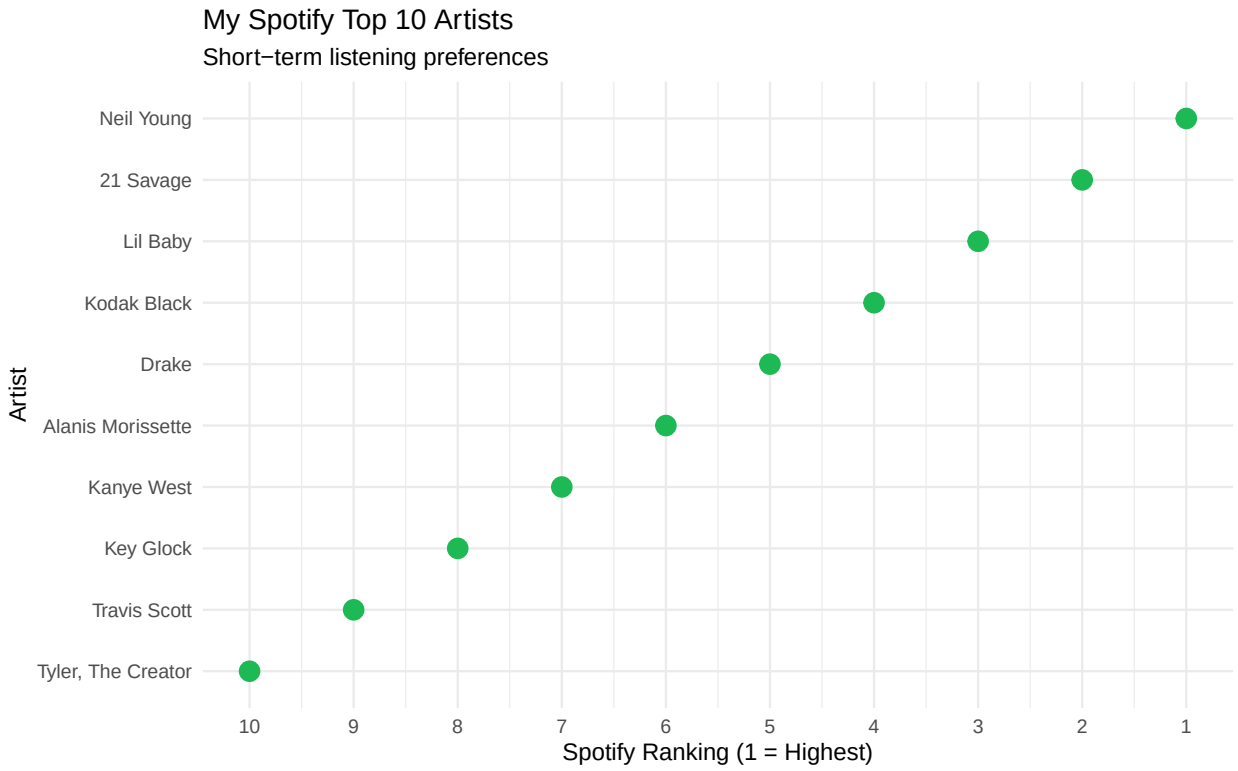


Figure 1: My Top 10 Artists: Short-Term Listening Preferences

This visualization illustrates the relative ranking of my most preferred artists during the short-term listening period.

4.2 Consistency Across Listening Periods

The next analysis examines how many listening periods each artist appears in.

```r
artist_consistency <- my_spotify_data %>%
  group_by(artist) %>%
  summarise(
    periods_present = n_distinct(period),
    .groups = "drop"
  )

artist_consistency %>%
  arrange(desc(periods_present), artist) %>%
  knitr::kable(
    caption = "Artist Consistency Across Listening Periods"
  )
```

Table 2: Artist Consistency Across Listening Periods

artist	periods_present
21 Savage	2
Drake	2
Kodak Black	2
Lil Baby	2
Neil Young	2
Rainbow Kitten Surprise	2
Tyler, The Creator	2
alt-J	2
2006WR	1
A Tribe Called Quest	1
Alanis Morissette	1
Beyoncé	1
Chappell Roan	1
Fred again..	1
Kanye West	1
Key Glock	1
Neutral Milk Hotel	1
Sade	1
TOOL	1
The Coup	1
Travis Scott	1
Zara Larsson	1

4.3 Visualizing Artist Consistency

```r
artist_consistency %>%
  count(
    periods_present,
```

```
  name = "number_of_artists"
) %>%
ggplot(
  aes(
    x = factor(periods_present),
    y = number_of_artists
  )
) +
geom_col(
  fill = "#1DB954",
  width = 0.6
) +
geom_text(
  aes(label = number_of_artists),
  vjust = -0.5
) +
scale_y_continuous(
  expand = expansion(mult = c(0, 0.15))
) +
labs(
  title = "How Consistent Is My Music Taste?",
  subtitle = "Artist overlap across listening periods",
  x = "Number of Listening Periods",
  y = "Number of Artists"
) +
theme_minimal()
```

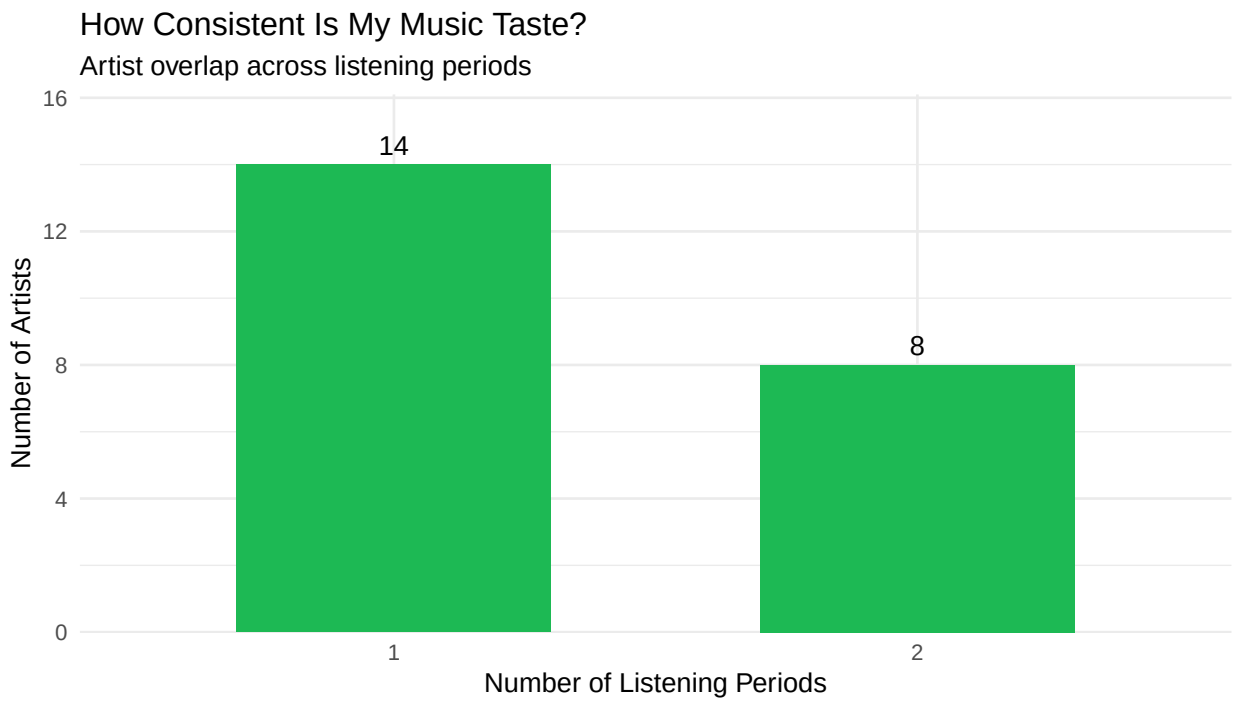


Figure 2: Artist Representation Across Listening Periods

Artists appearing in all three periods may represent more enduring musical preferences, whereas those appearing in only one period may reflect more time-specific interests.

4.4 A Heatmap of Listening Preferences

To visualize the overlap between listening periods, I created a heatmap showing which artists appear in each period.

```
heatmap_data <- my_spotify_data %>%
  complete(artist, period)

ggplot(
  heatmap_data,
  aes(
    x = period,
    y = artist,
    fill = !is.na(rank)
  )
) +
  geom_tile(color = "white") +
  geom_text(
    aes(
      label = if_else(
        is.na(rank),
```

```
      "",
      as.character(rank)
    )
  ),
  size = 3
) +
scale_fill_manual(
  values = c(
    "FALSE" = "#F1F1F1",
    "TRUE" = "#1DB954"
  ),
  labels = c(
    "Not in top 10",
    "In top 10"
  ),
  name = "Artist Presence"
) +
labs(
  title = "My Music Preferences Across Listening Periods",
  subtitle = "Numbers indicate Spotify artist rankings",
  x = "Listening Period",
  y = "Artist"
) +
theme_minimal() +
theme(
  axis.text.x = element_text(angle = 20, hjust = 1)
)
```

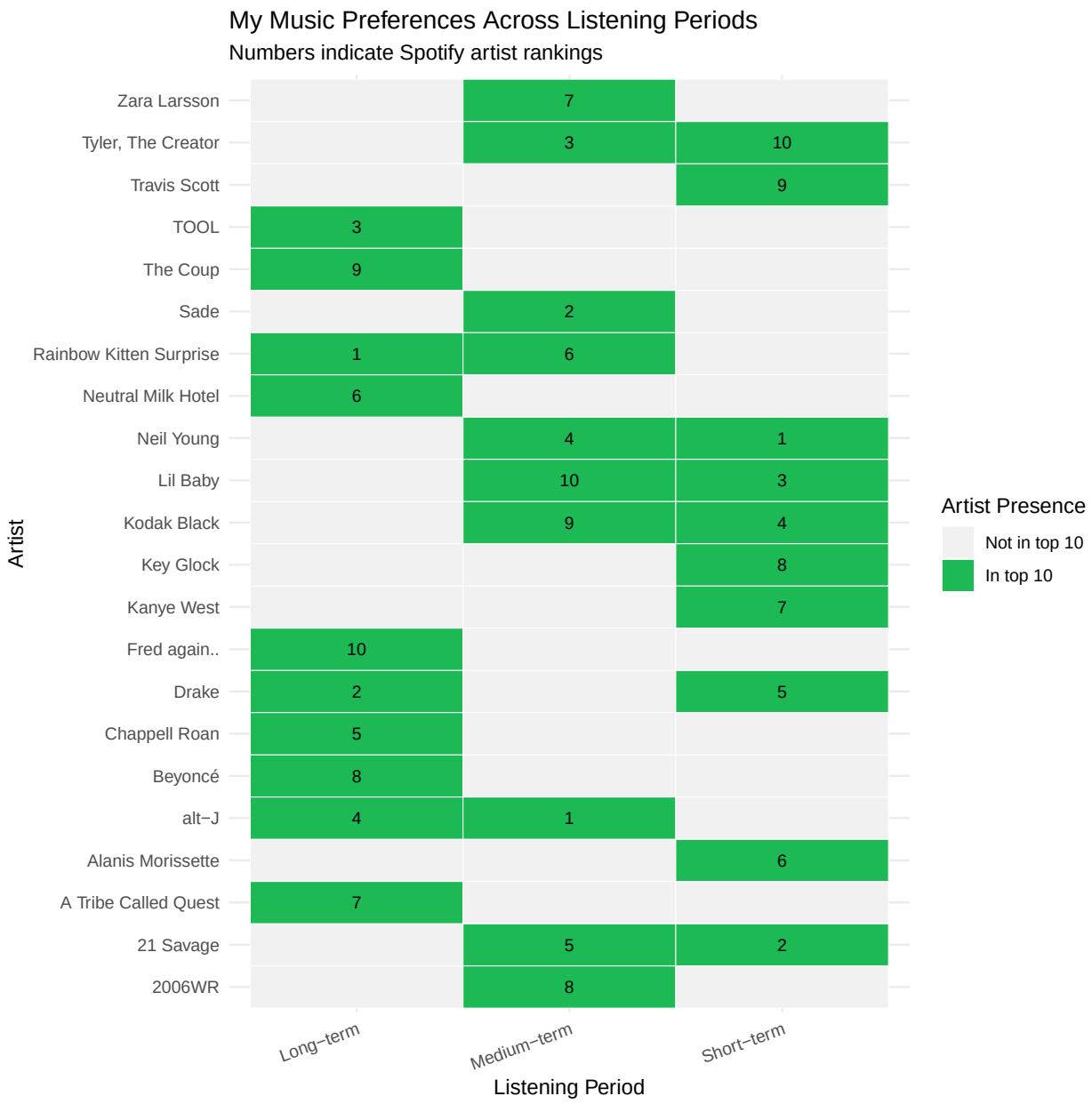


Figure 3: Artist Rankings Across Spotify Listening Periods

This heatmap illustrates which artists appear consistently across the three listening periods and which are concentrated in particular windows.

4.5 Recent Musical Preferences

Finally, I identify artists appearing in my short-term top ten but not in my long-term top ten.

```
recent_favourites <- short_term %>%
  anti_join(
```

```
    long_term,
    by = "artist"
  ) %>%
  select(rank, artist)

recent_favourites %>%
  knitr::kable(
    caption = "Artists Unique to My Short-Term Top 10"
  )
```

Table 3: Artists Unique to My Short-Term Top 10

rank	artist
1	Neil Young
2	21 Savage
3	Lil Baby
4	Kodak Black
6	Alanis Morissette
7	Kanye West
8	Key Glock
9	Travis Scott
10	Tyler, The Creator

These artists may represent more recently prominent listening preferences.

5 Discussion

This exploratory analysis demonstrates how personal listening data can be collected, organized, and visualized using R.

By comparing artist rankings across three time windows, the analysis provides insight into the consistency and variation of my musical preferences.

The findings also demonstrate the value of data visualization for identifying patterns that might be less apparent from individual rankings.

6 Limitations

Several limitations should be considered.

First, Spotify's rankings reflect calculated listening affinity rather than exact streaming counts.

Second, the three listening windows overlap, meaning that differences between them cannot be interpreted as changes across independent historical periods.

Third, the analysis includes only the top ten artists per period. Artists outside these rankings are not represented.

Finally, genre information was unavailable through the API response, preventing the originally proposed analysis of genre preferences.

7 Conclusion

This project provided practical experience with API authentication, data collection, data frames, data manipulation, and visualization using R.

It also demonstrated how computational methods can be used to investigate everyday patterns of cultural consumption.

Future analysis could incorporate extended Spotify listening history to examine musical preferences chronologically and investigate changes over multiple years.